

Ex.No.4: OFDMA Resource Block Allocation

Objective1:

Program:

```
clc; clear; close all;

% System parameters
bandwidth = 10e6; % 10 MHz
rb_size = 180e3; % 180 kHz per RB

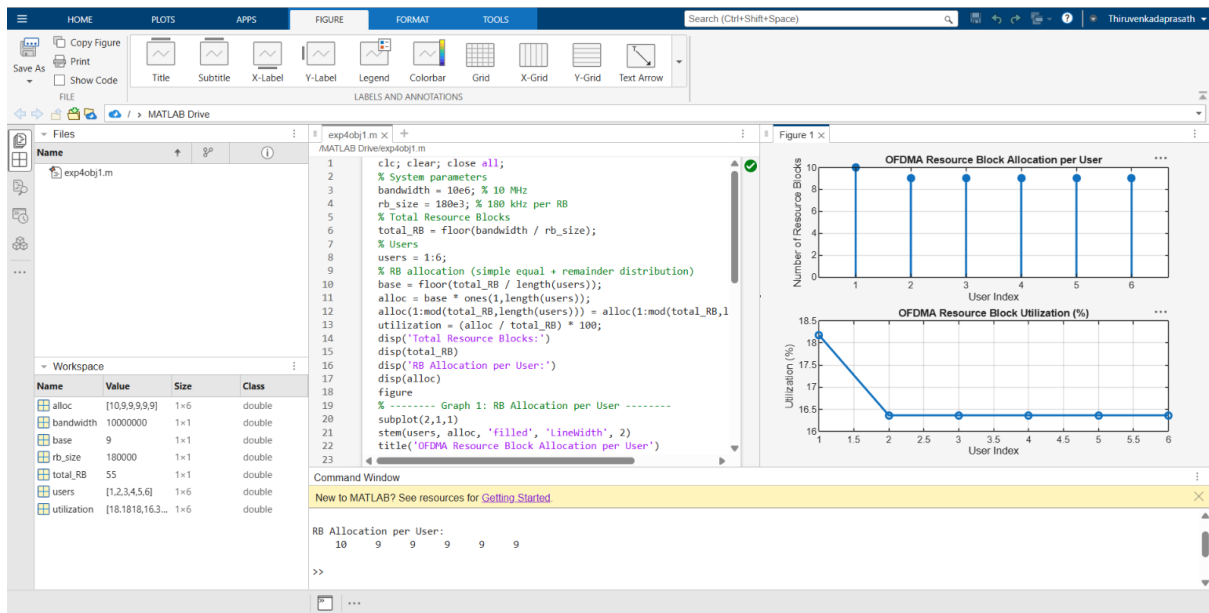
% Total Resource Blocks
total_RB = floor(bandwidth / rb_size);

% Users
users = 1:6;

% RB allocation (simple equal + remainder distribution)
base = floor(total_RB / length(users));
...xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on

% ----- Graph 2: RB Utilization -----
subplot(2,1,2)
plot(users, utilization, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```

Output:



Objective2:

Program:

clc; clear; close all;

% Total system resource blocks

total_RB = 50;

% Number of active users

users = 1:5;

% RB allocation to users (example scenario)

alloc = [12 10 9 8 11];

% Resource Block Utilization (%)

util = (alloc / total_RB) * 100;

disp('Resource Block Utilization (%) per User:')

...xlabel('User Index')

ylabel('Number of Resource Blocks')

grid on

% ----- Graph 2: Utilization (%) -----

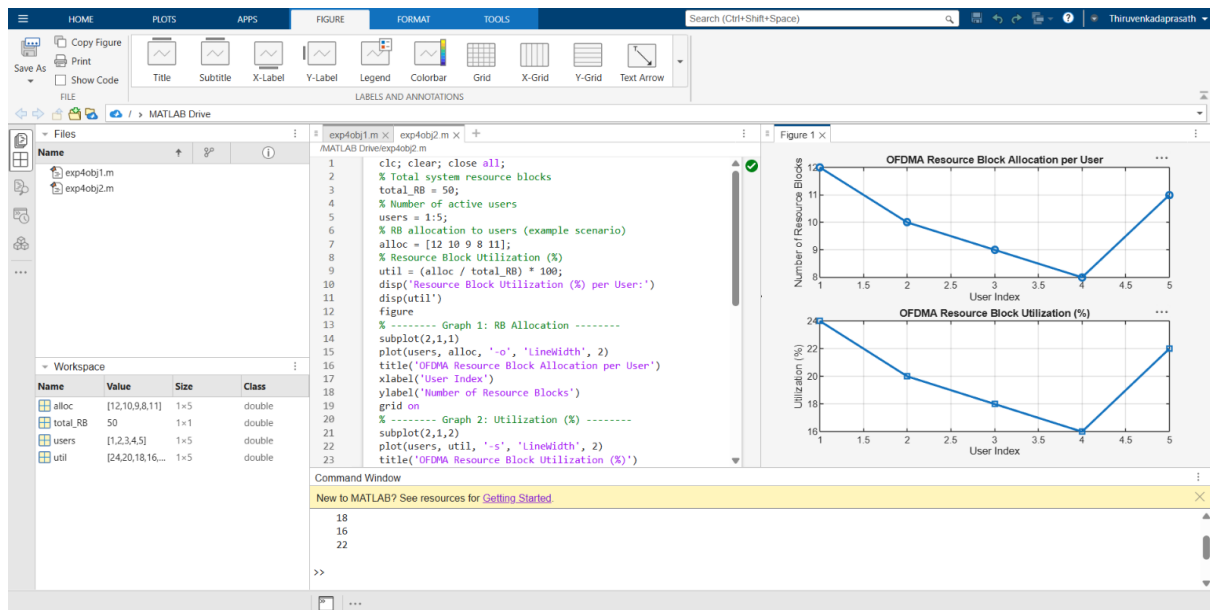
subplot(2,1,2)

```

plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

```

Output:



Objective3:

Program:

```

clc; clear; close all;

% System parameters
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
users = 1:5;

% Allocated Resource Blocks per user
alloc_RB = [12 10 9 8 11];

% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.

```

```
% Throughput calculation (Mbps)
```

```
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
```

```
...xlabel('User Index')
```

```
ylabel('Throughput (Mbps)')
```

```
grid on
```

```
% ----- Graph 2: RB vs Throughput -----
```

```
subplot(2,1,2)
```

```
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
```

```
title('Resource Blocks vs Throughput Relationship')
```

```
xlabel('Allocated Resource Blocks')
```

```
ylabel('Throughput (Mbps)')
```

```
grid on
```

